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I 001 INSTRUMENTATION AND CONTROL

1. GENERAL

This Technical Specification covers the general requirements for instrumentation and control (I & C).

In addition to the requirements of this equipment, all equipment offered to meet the requirements of I & C shall also comply where appropriate with other technical requirements of this Document.

A consistent I & C philosophy shall be applied throughout the plant and shall be implemented in terms of range of equipment exhibiting a minimum diversity of type and manufacture. The objective shall be to standardize all measurement and control equipment throughout the plant in order to rationalize operation, maintenance and spares holding.

Generally equipment shall be supplied from one composite range of measurement and control equipment as marketed by a reputable manufacturer of international standing and shall have minimum of 2 (two) years successful commercial operation use on similar projects to the effective date of this contract. Detail of similar equipment and installations shall be provided with the Proposal. Where particular equipment is not included in the composite range of the manufacturer thus necessitating diversity of supply, the equipment shall be subject to the approval of the Owner.

The equipment shall be of modern compact design incorporating the latest developments in proven technology.

All sizes and ratings specified in text or on drawings are the minimum reference requirements. If larger sizes or ratings are required in accordance with the Contractor's calculations or equipment characteristics, they shall be provided by the Contractor at no additional cost.

Any data collecting for the calculation or design purposes are the contractor's responsibility to conduct.

2. CODES AND STANDARD

The Contractor shall comply with the requirements of Part 4, subsection 2.5 “Codes and Standards” which the following are specifically applicable to the design, manufacturing and testing of the equipment included in this Technical Specification.

ANSI C37, 90a (IEEE 472)	- Surge Withstand Test
ANSI MC96.I	- Temperature Measurement Thermocouples
ANSI B31.1	- Code for Pressure Piping
ASME PTC 6	- Test Code for steam Turbine
ASME PTC 19.3	- Temperature Measurements
ASME PTC 19.5	- Application, part II of fluid meters
ICEA S-19, Part 6	- AC test Voltage Requirements
IEC 60050	- International Electro Technical Vocabulary
IEC 60051	- Direct acting indicating analogue electrical measuring instruments and their accessories.
IEC 60309	- Plug, socket Outlet and couplers for Industrial Purposes
IEC 60478	- Stabilized Power Supplies, - DC Output
IEC 60529	- Protection provided by Enclosures
IEEE 5 td 9I a	- Graphic Symbols for Logic Diagrams
ISA S5.1	- Instrumentation Symbol and identification
NEMA CC-1	- Electric Power Connectors
SAMA RC 22- 11	- Functional Diagramming of I&C Systems
BS I042	- Method of measurements of fluid flow in closed conduits
ISO 7919	- Vibration
ISO 5725-1	- Accuracy of measurement methods and results

3. ENVIRONMENTAL REQUIREMENTS

3.1 General Requirement

I&C systems furnished under this Technical Specification shall be suitable for operation in a power plant near the sea-coast in the tropical climate. I&C equipment shall be designed for continuous operation even if the air conditioning system provided for area (room) where the control equipment is located is not in service.

3.2 Degree of Protection by enclosure

All I&C panels including local mounted instrument panels, local mounted instruments and their junction boxes shall have the following degree of protections as defined by IEC 60529, unless otherwise specified.

- a. IP 31 - For indoor in an air conditioning room
- b. IP 51 - For indoor in an enclosed room
- c. IP 52 - For indoor away from fluid piping and tanks
- d. IP 54 - For other indoor locations
- e. IPW 55 - For outdoor panels
- f. IPW 65 - For local junction boxes, local mounted instruments including pressure switches, level switches, temperature switches, vibration pick up etc.

3.3 Central Control Room and Electronic Equipment Room

The computer or electronic equipment shall be installed in a Central Control Room and Electronic Equipment Room which shall be an air conditioned room, with fully redundant air conditioning equipment. However, for those periods of time when the air conditioning unit is out of service, the equipment shall operate satisfactorily without forced air until a maximum temperature of 35°C is reached external to the cabinets.

The design requirements for ventilation and air conditioning and lighting of these room, shall be as described in Part 4, Technical Specification M-421 “Ventilation and Air Condition System” and E-270 “Lighting”.

4. DOCUMENTATION

4.1 Logic Diagrams

The Contractor shall furnish Control Logic Diagrams for complete Instrumentation & Controls. Control Logic Diagrams shall be in accordance with SAMA (Scientific Apparatus Manufacturer's Association), Standard RC 22-11 and with ANSI Y 32.14.

A complete write-up of the operation of the controls shall accompany the logic diagram. The logic diagram and write-up shall together present a complete description of the operation of the control systems during normal operation, start-up, shutdown, plant trip, and loss of an auxiliary such as a fan or pump.

4.2 Instrument Location Drawings

The Contractor shall furnish drawings showing the location of the various sensing points, sensing lines, and the instruments measuring the variables. The routing of sensing lines to Local or Panel Mounted Instruments together with details of sleeves, floor stands, embedded plates, and panel connection and identification on the general arrangement and piping drawing shall also be provided.

Sensing points and thermo wells shall be shown on the piping drawings together with notes that apply to instrumentation works.

All pipes shall be identified by their own pipeline numbers and the elevation shall be shown. The instrument taps shall be identified by the instrument numbers they serve.

All instruments shall be identified by their own instrument numbers. Where an instrument is panel mounted, its panel number shall identify the panel and a table on the drawing shall identify the instruments on the panel. The elevation and drawing column coordinates for the panel shall be given in the table. The panels shall be numbered in accordance with Part 4, Sub-section 2.8.

Solenoid valves, which are physically separated from the process valves they control shall be shown together with the control airline from the solenoid valve to the process valve.

4.3 The Attached Drawing

The attached drawings are shown as minimum requirements and for reference only.

- 1222-00-I-55-C-001 DCS configuration Block 2
- 1222-00-I-55-C-002 DCS Configuration Block 3
- 1222-34-I-55-C-003 DCS Configuration Block 4
- 1222-00-S-07-UCB-001 Central Control Building Floor Plans
- 1222-00-I-70-C-005 Overall Tripping Logic Diagram

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5. UNIT OF MEASUREMENT

Unit of measurement shall be in accordance with Part 4, Sub-section 2.15.

6. DESIGN CONCEPT

6.1 I & C Design Concept

A fully comprehensive control, monitoring, measurements and protection system shall be provided for safe and efficient operation of the GTG, HRSG, STG, auxiliaries, balance of plant and electrical system. The Plant Control System shall be designed generally on the basis of a functional distribution philosophy. The technology shall be microprocessor based using proven hardware and software. To meet the design requirement, the plant control system should be engineered by related DCS manufacturer.

The overall I & C system shall be designed to ensure the following as minimum:

- Maximize safety for the equipment and plant personnel
- Safe, reliable and efficient operation of the plant under all operating conditions.
- Maximize availability and efficiency of the plant
- Consistent instrumentation and control philosophy including a high degree of simplicity, operability and maintainability
- Ability to operate under all specified conditions (start-up, normal operation, disturbances and shut down) automatically and/or by remote manual control from the relevant control room

- Malfunction of any component or loss of supply will lead to safe condition (fail-safe)
- Provide accurate and reliable information to the plant operators and management
- Apply most modern control system
- Uniformity of I & C equipment to be supplied for maintainability

Adequate redundancy and back-up control shall be provided where necessary so that failure of single component will not lead to an abnormal condition, degrade the performance or shorten the life of the main plant equipment and electrical system. Redundancies shall be ensured in terms of providing redundant sensors, control equipment and communication equipment wherever necessary. 2 out of 3 philosophy shall also be adopted for critical control loops and interlocking system. For safety reasons, the process control for critical system shall be independent system and safety certified to SIL3 or AK6, provided with own monitoring and controlling system. The system shall be linked to DCS via redundant high speed data highway.

6.2 Overall Plant Control and Monitoring Philosophy

The overall central operation and supervision of the gas turbine generating unit, heat recovery steam generator (HRSG), steam turbine generating unit, auxiliaries, balance of plant and associated electrical system shall be coordinated from a central control room (CCR).

6.3 Automation

The plant shall be designed to operate with minimum number of staff consistent with safe and efficient operation. All the units, whether on open cycle or combined cycle operation, shall be controlled and supervised primarily from the CCR. All unit start-up, shut down, load dispatching, electrical switching functions and supervision of common services shall be exercised from the CCR. The extent of remote control and level of automation shall be such that one operator shall be capable of supervising and controlling the day to day running of each unit from the CCR with little assistance from local operators.

The plant shall be designed to operate all unit start-up and shut-down automatically, supervising from CCR.

6.4 Operation Mode

The plant shall be operated at following basic mode.

6.4.1 Open Cycle Operation Mode

Only the GT and its auxiliaries are in operation and the exhaust gas is going through the bypass stack using diverter damper.

6.4.2 Combined Cycle Operation Mode

The nominal operating of combined cycle mode with both GT and ST are in operation and exhaust gas is going through HRSG and exhaust stack.

6.4.3 Mode Transfer

Transfer from open cycle operation to combined cycle operation:

- The plant shall be capable of changing from open cycle operation to combined cycle operation automatically without reducing the GT load, using the exhaust gas diverter damper and turbine bypass valves

Transfer from combined cycle operation to open cycle operation:

- The plant shall be capable of changing from combined cycle operation to open cycle operation automatically by operating the exhaust gas diverter damper. Then the turbine bypass valves will open, the ST steam valves will close and the ST will be tripped.

6.4.4 Load Operation Mode

The combined cycle plant shall basically be designed for base load operation and alternatively two shift operation.

6.4.5 Fuel Transfer Mode (not applicable)

The plant shall be capable of changeover from fuel gas to distillate oil and vice versa on load with no bumpless.

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6.5 Control under abnormal condition

6.5.1 Grid isolated operation

Refer to Part 4 Section 3 I-100 "Plant Control System"

6.5.2 Bypass operation using by the bypass dumper

Refer to Part 4 Section 3 I-100 "Plant Control System"

6.5.3 Bypass operation using the turbine bypass valve

Refer to Part 4 Section 3 I-100 "Plant Control System"

6.6 Control System Philosophy

6.6.1 Distributed Control System (DCS)

The Control System shall be strictly segregated to minimize the possibility of a common mode failure.

All control and supervisory functions shall be centralized in the CCR and arranged so that the Plant can be conveniently controlled and supervised by a single operator.

The control system adopted for the Plant Control and supervision shall be from a fully functionally distributed micro processor based Distributed Control System (DCS) type. It is preferred that the data/signal transmission from the plants/equipment to the controller and supervisory system and vice versa shall be by a redundant serial data link installed throughout the plant. This link shall preferably be by fibre-optic cable but short runs may use coaxial cable.

6.6.2 Operator Interface

In general all manual and automatic control functions shall be initiated and supervised remotely from the Operator station located in the CCR. However, a small number of exceptions to this shall be permitted in order to make the task

of the single operator easier. A certain class of individual controls are not required in the main control room, but may be controlled only by the sequencing system. During commissioning and maintenance these items may be controlled and tested from the 380 V motor control center. This group includes those drives fulfilling the following criteria:

- Drives not subject to plant item interlocks.
- Drives which if their associated subgroup loses its control facilities the plant can carry on operating and also be shut down safely without the state of the drive having to be altered (for example, the plant item may be allowed to remain running during shutdown).

Local trip pushbuttons for individual drives shall not in general be provided because of the probability of unwanted operation, reliance being placed on the item interlocks to ensure plant protection under fault conditions. However, local emergency trip buttons shall be provided for the Steam Turbine unit. The 380 V drives shall be equipped with local stop pushbuttons only if the mechanical arrangement of the drive is considered a personnel safety hazard.

Plant data required for the normal operation and supervision shall be presented to the operator on the local control panels and in the CCR. For local control see Section I-002 Local Control Panels, and for control from the CCR see Section I-005 DCS

6.6.3 Control System Failure

The control system shall comply with the following general failure criteria.

- (i) No single fault shall cause the complete failure of the control system.
- (ii) No single fault shall cause ST generator, HRSG or steam turbine protection system to spuriously operate or cause the protection system to become inoperative.
- (iii) The grouping of the control functions into system modules shall be arranged such that failure of any one module will only partly degrade the control of the overall system. Such degradation shall always be manageable by operator intervention.
- (iv) The control system shall be structured to reflect the redundancy provisions of the plant so that no single fault within the control system can cause the failure of the duty equipment and at the same time cause

the standby equipment to be unavailable.

- (v) If as a result of a control system fault a plant item or control function cannot respond to its controls at the item level (i.e. Remote Manual Control Level) then that item or control function shall be caused to fail to safe condition approved by the Owner.

To meet the above failure criteria the I & C system shall incorporate self checking facilities so that internal faults can be detected within the I & C system itself prior to any resulting disturbance to the process. In addition, the protection and safety systems shall include monitoring of measurements as well as self-checking and adequate test facilities.

To meet the stringent failure and self-checking requirement for the I & C system, measurement redundancy shall be provided for all critical plant parameters.

The measurement for the functions of control, indication and protection may be combined provided that the integrity of measurement is ensured by adequate channel redundancy.

For control and indication of critical parameters where duplex measurement is considered appropriate, the control system shall automatically select the higher, lower or the average as the duty signal in accordance with the safe operation of the plant. Provision shall be made to over-ride the automatic selection by selecting either measurement and for continuous difference discrepancy monitoring.

For protection purposes, basically multi-channel measurement shall be provided incorporating a 2 out of 3 voting philosophy. The Contractor shall provide a diagram indicating systems using single and multi channel for approval by Owner. A redundant data transmission bus shall also be provided. Wiring from redundant control devices shall follow different routings.

6.6.4 Plant Operation

The power plant is required to be operationally highly flexible so that it shall satisfy the full range of operating regimes likely to be demanded during its operational lifetime. The unit shall be capable of two shift operation, participating in frequency control of the grid and load regulation over its full load range. In order to minimize the burden on the unit operator all control

functions within the operational load range shall be fully automated. The unit start-up from overnight shutdown and from shutdown states of up to 60 hours shall be fully automated with the exemption of a degree of manual preparation of the steam cycle plant. Also, unit rundown is to be similarly automated. However, start-up and shutdown control shall also be possible by the separate initiation of the various functional group or sub-group start/stop programs. The extent of the automation and supervisory functions shall enable a single operator to exercise management over all normal block operations. A single pushbutton start shall be provided. For cold block start-up the availability of an auxiliary operator may be assumed to assist with local plant supervision and operation.

The combined cycle plant shall be fully co-ordinated and provision made for the remote load dispatch control from a regional grid control center. The requirements for plant co-ordination are specified under this specification.

A functional group control strategy is to be adopted for the control of all the principal plant items and their auxiliaries. This will form the basis for the arrangement of the controls on the operator console at the CCR. The control system shall be functionally distributed, highly modular and arranged in a hierarchical structure closely reflecting the functional grouping of the plant to be controlled.

6.6.5 Drive Control Level

Every remotely operated drive or circuit breaker shall be controlled by means of a drive control device. To this device all orders i.e. from the operator station, the release button on the operator station and from the subgroup automatic system, if any, shall be connected. Furthermore, all interlocks and the feedback signals of the drive or switchgear shall be connected to control this control device. After the processing of these signals the control device shall issue the order to the switchgear via interposing relays.

The interposing relays shall be housed in the relay room of the switchgear or integrated in the switchgear housing and are to be considered as part of the logic control system. Solenoid valves operating at the same voltage level as the drive modules can be energized directly from the drive control module.

Additional local start/stop pushbuttons shall be provided in cases where local operation of the plant is necessary. In all such cases the local controls shall act

only if selected by a group or individual selector switch, located on local site.

Group selector switches shall be provided where a large number of local start/stop pushbuttons are required.

In cases where it is necessary, for safety reasons, to provide a local emergency off pushbutton, the pushbutton shall directly interact with the switchgear. The emergency push button shall be provided with suitable interlocking device in order to avoid automatic restarting.

The priority of protective signals over manual or automatic commands must be assured.

If, due to motor protection, dry-run protection with pumps, etc., a switch trip occurs, immediate switching on must not be possible before the switch tripping signal has been acknowledged.

For all low voltage drives except for actuators fed by the various 380 V busbars, provisions shall be made in each individual command logic to switch-off the motors in the case of a short busbar voltage drop below a determined value. However, the motors shall restart automatically, preferably in sequence if the voltage restores within certain time.

Any voltage drop beyond the above delay period shall lead to definitive switch-off command for all drives connected to the corresponding busbar. Binary signals from under voltage relays installed for each half busbar shall be provided by the Contractor.

In addition, great importance is attached to the fault annunciations per drive e.g:

- discrepancy signals, that means an annunciation, if the position of a drive contract is in opposition to a given command.
- tripping signals in the case of faults in the switchgear (motor overload protection etc.) active protective interlocks, local switching, and fuse failure.

In the event of any switch tripping, a group signal must be passed to the alarm annunciation system.

In addition, the tripping signal must be displayed individually in the appropriate operation element in the control room by a flashing signal. An individual steady signal shall also be attached to signaling and monitoring of the control modules, e.g. correct plug-in connection, failure of any fuse etc.

6.6.6 Sub-Group Control Level

Sub-group controls are to be provided for start-up, operation and shutdown of associated functional units and drive groups. The controls may be designed either in the form of logic controls or in the form of sequence controls. In the case of logic controls, the issue of command depends solely on the correct setting of certain process criteria. In the case of sequence controls, the command sequence takes place in steps and is monitored for time, taking into account incoming process criteria in a predetermined sequence and in a fixed scanning time.

The steps of the individual programs and missing criteria shall be displayed on the HMI in report fields (step and criteria indicating). See Section I-200 “Plant Management System” for details of sequence control monitoring.

Any faults in the control system which could cause blockage of the control sequence must be confined to one control sequence. For this reason an autonomous design is required for each subgroup control.

All functions performed by automatic system must be capable of being implemented as individually remote-controlled operations via command modules.

6.6.7 Group Control Level

Group control level shall be provided for start-up, operation, shutdown; automatic start-up of the standby devices of the complete units (feed water pump, turbine, etc.).

Group control level shall also include change of the structure and parameters in closed control loops.

6.6.8 Plant Coordination Control Level

The fourth level, which stands at the apex of the hierarchical triangle, performs the overall plant co-ordination. It incorporates, for example, the

master load control function which would regulate the load generated by the generators to the demand set point value. In general for automatic start-up and shut-down control the sequential initiations of the functional group sequences would be generated at this level.

This mode is the highest level in the control system architectures to conduct unit co-ordination control to perform APS (Automatic Plant Start-up and Shutdown) and Automatic Power Regulation (APR) with associated peripheral devices/system.

6.6.9 Hardware Arrangement Philosophy

The physical and electric segregation of the control system hardware shall directly reflect its functional hierarchy. The control elements associated with a particular functional group or sub-groups at the same hierarchical level shall be grouped together and segregated from all other control element groupings. It is not permitted to implement several subgroup controls within a common hardware. Further, the drive level hardware shall be separate from subgroup hardware and each drive shall have its own control module. Controls elements associated with a functional group shall be capable of being tested and operated independently of other functional groups. Within the bounds of each functional group the lower hierarchical control functions are to be capable of independent operation from those at higher levels.

However, at the functional group level a small number of intergroup communication signals are permitted in order to improve the system integrity in the event of failure at the coordination level. This inter functional group communication is to be indicated on the plant coordination control block diagrams.

The hardware at the plant item control level shall be discrete in nature and all signal communication hard wired. At the higher levels of the control hierarchy, the hardware may become more freely configurable and at the plant coordination control level this is considered desirable. Reconfiguration shall be possible by the end user and all necessary facilities shall be provided. These facilities shall be designed for the use of engineers without extensive programming experience.

The control equipment shall be housed in cubicles, each cubicle or suite of cubicles being dedicated to a particular functional group or sub-group.

For the purpose of reconfiguration stated above a combination of an on line diagnostic station shall be provided for the control system of each combined cycle block.

The Diagnostic and Engineering Stations are the central working place for the maintenance personnel and allow the monitoring of and dialogue with the entire control and monitoring system.

The Diagnostic Station shall perform at least the functions set out in Part 4 Section 3 I-005 “Distributed Control System”.

Each diagnostic and engineering station shall also include VDU, keyboard, and printer.

6.6.10 Interlocks, Protection

To protect individual units or parts of the plant, interlocks are to be formed in accordance with process criteria, which, depending on their functions, can be either active or passive.

Active interlocks shall automatically disconnect units or parts of the plant before they reach a critical operating condition. In addition, such dangerous conditions must be immediately indicated to the operating personnel by means of the alarm annunciation system and shall lead to a print-out by the computer.

Passive interlocks are intended to prevent operational errors or wrong commands from being carried out in the event of selective faults in the automatic control.

For that reason active and passive interlocks must normally not be capable of being switched off operationally from the control room.

Large and expensive parts of the plant and units shall be protected by requirements safety circuits (refer to 6.6.3 control system failure) must be employed.

6.6.11 Local Instrumentation and Control

Controls provided local to the plant or on switchgear shall in general be excluded from the scope of the above operational control levels. In general, local controls shall only be provided for the purpose of plant testing under the

following circumstances:

- (i) Where no direct remote item control is provided in the CCR.
- (ii) Where the drive is not subject to any plant interlocks.

Local control system for major ancillary plant such as water treatment plant etc. shall be microprocessor based DCS or programmable logic controller.

Local direct reading instrumentation and controls not forming part of the central control and supervision system shall be provided as follows:

- (i) Local control and indications shall be provided for the following general functions:
 - (1) For general plant start-up preparation.
 - (2) For local plant systems which are not required for automatic start up and require no direct remote supervision or control.
 - (3) For the indication of the operational state of the plant where remote monitoring instrumentation is not required.
 - (4) For local plant testing and maintenance.
 - (5) Where required for the convenient checking of other instrumentation (e.g. a pressure gauge to check the operation of a pressure switch, etc.)
- (ii) For general start up and maintenance all necessary local instrumentation shall be provided including at least the following

Pumps	: Suction and discharge pressure gauges Gland sealing pressure gauge (if applicable)
Strainers and Filters	: Differential pressure gauge(s)
Heat Exchangers	: Temperature upstream and downstream, both tube side and shell side.
Lubrication and Cooling Systems	: Pressure gauges indicating system pressure at appropriate locations to enable correct system

functioning to be confirmed.

- (iii) Minor local automatic control loops normally of the self acting type (requiring no remote monitoring or operator intervention within the normal unit load cycle) may be packaged within the supply of the associated plant items.

Where indications and controls are not directly linked to the remote control sequence and interlock system, local instrumentation may be adopted subject to the approval of the Owner.

Local indication instruments shall only be provided where continuous supervision is unnecessary and where local indications are required for local plant commissioning and adjustment. In all cases where continuous monitoring is required, a transducer and remote monitoring shall be provided.

Local control shall only be provided in cases where under normal circumstances no remote control is necessary within the normal load range of the units. Motorized actuators with local control shall be provided for large valves where hand operation is inappropriate. All motorized valves with integral switchgear shall be provided with local control for test purposes with a lockable local/remote control selector switch. All locally mounted instruments shall be located in readily viewable and accessible positions.

6.7 Power Supplies and Cabinets

6.7.1 Control System Power Supplies

Each power supply system for control system shall include both “main” and “back-up” equipment, each being capable of assuming the main or the back-up function. Upon failure of the main equipment, the load shall automatically transfer to the back-up equipment without interruption to the power output of the power plant. Alarm contacts shall be provided to notify the operator of failure of any power supply or regulating equipment. Provision shall be included for manual switching from one power supply to the other for test and maintenance purposes.

Each power supply bus system shall have minimum capability of 125 % of the maximum load.

Indicating light emitting diodes (LED) shall be provided on the front door of

the termination cabinet for each power supply bus system to indicate which power supply unit is in service, in stand-by condition, or failed.

Power supplies shall be located to minimize deterioration for heat and high ambient temperature increase in the system and termination cabinets. If power supplies are located at the bottom of a cabinet, deflectors and/or exhaust ducts shall be provided.

The nameplate shall be furnished by the Contractor for all power supplies both front and rear. The detailed requirement for nameplates required, refer to Part 4, Technical Specification I-002 “Local Control Panels”.

6.7.2 Control System and Termination Cabinets

The system cabinets shall contain the hardware required to develop the control logic and interface with the operator and other system peripherals. The termination cabinets shall contain the hardware required to terminate the field wiring, to buffer the field input signals and to provide heavy-duty contacts for output signals.

For requirements applicable to both types of cabinets, refer to Part 4, Technical Specification I-002 “Local Control Panels”, when applicable.

6.8 I & C Cabling Concept

6.8.1 I & C Cable

A Distributed control system shall be provided to reduce the quantity for I & C cabling.

All cabling between the digital system including RIO and their peripherals shall be optical fiber cables or co-axial cables To reduce the quantity of I & C cabling, marshalling boxes may be installed. The cabling between field equipment and marshalling boxes shall be individual cables; the cabling between marshalling boxes and other panels including I/O cables shall be multicore cables or prefabricated cables.

6.8.2 Grounding of Control System

Each instrument and control circuit ground shall be a single insulated copper

wire not less than 2.5 mm² which shall be run (independently from the safety grounding system for power cable) to the ground bus and fastened thereto with a suitable terminal lug. The cases of all I & C panels, recorders etc are considered to be effectively grounded through the panel structure.

6.9 Signal Interface with other system

6.9.1 Serial Interface

In the case of long distance communication between digital equipment such as plant Bus or Communication Bus, serial interface shall be provided. In this case fibre-optic cable shall be used.

6.9.2 Parallel Interface

In the case of short distance communication between digital equipment and their peripherals, such as LAN, parallel interface shall be provided in this case.

6.9.3 RIO Interface

In the case of connection between DCS and local instrument or controls, RIO shall be used for saving the quantity of cables.

6.9.4 Gate Way Interface

If a different type of hardware from that of the DCS is provided, a gate way controller shall be provided for the communication between them using fiber optical cable.

6.9.5 Hardwired Interface

Hardwired cable shall be provided for following I & C wiring.

- Connection between field equipment and related local panels.
- Input wiring for event recorder system
- Input and output wiring from RIO to field equipment or local panels.

7. DESIGN REQUIREMENTS

7.1 Control System Design Requirement

A consistent instrumentation and control philosophy shall apply throughout

the plant and be implemented in terms of a range of equipment exhibiting a minimum diversity of type and manufacture. The Contractor shall ensure that the design, supply and implementation of the plant instrumentation and control functions, with only agreed exceptions, shall be the direct responsibility of a single control system supplier, who shall either be the Contractor or his appointed Instrumentation and control Sub-Contractor.

The Control system design standards set out in this Specification shall apply throughout the plant; however where they relate to the manufacturer's standardized control systems it will be considered that the supply options offered meet these standards. It shall be understood that such option will only be considered for systems which are subject to series production and whose reliability, operability and serviceability can be clearly demonstrated by the Contractor. Even in such cases the systems offered shall comply with the general scope of this Specification and the requirements which are considered specific to this project. Where the Bidder standardized systems do not comply with the Specifications, he shall state the exceptions in the schedule provided and include with his tender sufficient technical details and justification to leave no doubt as the nature of his offer.

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To meet the stringent response demands and the operational requirements a highly flexible control regime is required with a choice of facilities to allow on-site optimization of the system to meet the actual response requirements at initial commissioning and during the later life of the station as the load patterns vary due to the development of the power system.

7.2 Primary Devices

7.2.1 Flow Elements

Flow metering shall be in accordance with BS 1042 "Methods of Measurement of Fluid Flow in Closed Conduits".

It is preferred that flow meters shall be of the differential pressure type. The working fluid shall not cause deterioration of meter material and associated parts. Flow meters shall have their outputs linearised. Accuracy shall be within 0.5 % of full- scale range.

Where flow is measured by differential pressure devices, (e.g. Venturi tube, averaging pitot tube ("Annubar") flow nozzle or orifice plate), materials of

construction shall be suitable for the particular service in which the device is located. Orifice plates and flow nozzles shall be of stainless steel. Every orifice plate shall have an integral tab or handle, which shall extend at least 75 mm beyond the pipe lagging, or flange lagging and shall have the following items permanently marked on the inlet face of the handle: orifice instrument number; orifice throat diameter; flange size, flange rating and "INLET". The identifying information shall be located on the outer 25 mm end of the tab or handle for easy identification.

Venturi Tubes and Flow Nozzles shall be provided in cases where low loss is required. These items shall also be designed in accordance with relevant standard.

The Contractor shall be responsible for verifying that sufficient length of straight pipe is provided on each side of all primary elements.

All measuring devices required, complete with all necessary condensation chambers, isolating valves, seal pots, flanges and connections in the pipework shall be provided.

Where necessary, flow metering shall be temperature and/or pressure compensated.

Differential pressure primary elements in hot liquid services shall be under sufficient hydrostatic head to prevent flashing at the lowest pressure point in the element outlet.

Differential pressure type flow meters shall be fitted as a minimum with 3-valve manifolds.

Where the environment is corrosive or may cause blockage of a tapping point, a continuous line purge system shall be provided.

Local flow meters and turbine flow meters shall be provided as required. Variable Area flow meter (if required) shall be only used for local flow indication.

7.2.2 Thermocouples, RTD's, Connection Head and Protection Wells

Thermowells and temperature measuring equipment shall be fitted in positions

to suit the operation and testing of the plant. Temperature measuring primary elements for distant reading shall be resistance thermometers or thermocouples.

7.2.2.1 Thermocouples

Thermocouples shall be ISA Type E (chromel/constantan) or type K (chromel/alumel). The limits of error are based on a reference temperature of 0°C and shall be accurate to ± 0.5 % of measured temperature.

Where a protection tube is required, the thermocouple wires shall be separated by ceramic insulators, which are encased in an alloy (nickel/copper alloy steel, ASTM 333 Grade 9) protecting tube.

Thermocouples specified, as “calibrated thermocouples” shall have a minimum three-point calibration, guaranteed to accuracy within ± 0.2 % of full range, and shall be furnished with individual certified calibration data.

An extension nipple of sufficient length to extend beyond any lagging and a connection head shall be furnished for each thermocouple assembly where necessary. Where thermocouples are installed in high temperature steam lines, thermocouple heads shall extend at least 200 mm from thermo wells. If removal of the primary element is restricted by the head-mounted transmitter, separate locally mounted transmitters shall be used. The base of the connection head shall be at least 75 mm clear in all directions of pipe lagging and shall be arranged to avoid excessive temperatures at the head assembly.

Mineral-insulated metal sheathed thermocouples shall be thoroughly dried out before installation, and the insulation resistance value shall be not less than 2 Mega ohms before the cold seal is made.

7.2.2.2 Resistance Temperature Detectors

RTDs shall be three-wire type with 100-ohm (at 0°C) platinum resistance elements to BS 1904 unless specified otherwise. They shall be certified to meet the requirements of the International Temperature Scale, calibrated to primary standard furnished with the constants of resistance versus temperature defined in the primary standard and guaranteed to an accuracy within $\pm 0.3^\circ\text{C}$ of this standard.

7.2.2.3 Connection Heads

Thermocouple terminal blocks shall be marked with polarity. RTD connection heads shall have properly labelled terminal blocks.

Terminal screws shall be resistant to heat and galling. A grounding terminal shall be provided in the connection head suitable for terminating cable shield drain wire. All connection heads shall be of cast iron and shall be complete with screwed cover caps providing a weather tight enclosure. A chain shall be provided to prevent loss of cap.

7.2.2.4 Filled Systems

Where filled system thermometers are used for local indication, they shall be of the direct mounted rigid stem type, but where a capillary is required it shall be armoured and its length shall not exceed:

- a) 4 metres when used for alarm or control purposes
- b) 10 metres for other purposes

7.2.2.5 Thermowells

Suitable thermowells or protective sheaths of approved design shall be provided for all types of temperature measuring elements to enable instruments to be replaced without requiring the plant to be shut down. Thermowells shall be installed in threadollets unless otherwise specified or approved.

Each thermocouple and temperature element shall be permanently identified with its assigned number as specified on the Instrument Data Sheets.

Wells shall have suitable clearances so that elements may be removed when operating at maximum temperature.

The temperature capability of each thermo well shall be stamped on the well and shall appear on the mill test reports (material certification and test reports). These reports shall be submitted to the Owner for review.

Thermowells shall generally be downstream of pressure instrument connections. A check thermowell shall be provided next to the thermowell for each remote reading temperature instrument, and plant efficiency check

thermowells shall be supplied where required. Each check thermowell shall be fitted with a removable plug which shall be threaded Rp 3/4 to BS 21.

Thermowells shall be of the quick response type and shall be of an approved material for the service. Thermowell insertion lengths shall be appropriate to the measurement process location.

Special care shall be taken in the design and installation of thermowells mounted in locations subject to movement, vibration and high velocities.

Thermowells shall be located in positions where there is sufficient space for removal of the instrument without interfering with adjacent equipment, and easily accessible for maintenance of the instruments fitted in them.

For reference, the manufacturer shall submit to the Owner all calculations that are required to check each type of thermo well design in accordance with ASME PTC 19.3, where applicable. The manufacturer shall state the response time to a step change in measured variable for each type of thermo well.

7.2.3 Switches

It is generally preferred that discrete points are developed in the DCS by level detection from analogue points. Where switches are approved by the Owner, they shall conform to the following clauses.

7.2.3.1 Switch Assembly:

All switch contacts shall conform to the following:

- Type.....The switch shall have dry contact. When atmospheric corrosion can be a factor, complete switching element shall be hermetically sealed.
- ActionTwo (2) electrically independent single-pole double-throw (SPDT). contacts corresponding to each set point . Repeatability will be ± 0.5 % of range or better.
- Rating5 Amps continuous at 220 V AC and 0.5 Amps inductive (minimum) at 250 V DC.
- SettingsIndividual “ON” and “OFF” points on a calibrated scale or dial visible and settable from outside of case.

Pressure switches shall be in accordance with the following:

C a s e:

- Type.....Water tight with a degree of protection of IPW 65, back flanged, front glass.
- Material and Finish Steel, synthetic lightweight material, or equal approved by the Owner. Finish shall be dull black or gray.
- ConnectionsBottom male pipe thread and back conduit hub, both suitable sized for the service. The wetted part of the pressure switch shall be suitable for intended service. Pressure part shall be 316 SS unless otherwise specified.

Measuring Element:

- Type..... Bourdon tube, bellows or diaphragm
- Material..... Above 70 bar - bored alloy steel
 28 - 70 bar - steel or alloy steel
 Below 28 bar- bronze or steel
- Range..... Provide standard range nearest to one specified to provide switch actuation at the maximum operating pressure or any adjustable point below.

Sensing elements shall be suitable for the ranges specified and shall also be capable of withstanding a minimum momentary surge pressure of 1.5 times full scale range without permanent damage or loss of calibration.

Accessories

- All pressure switches on steam service and water service over 65°C shall be complete with siphons. Pulsation dampeners shall be furnished

where required per manufacturer's recommendation, to limit the amplitude of pressure variations to suit the service condition of each pressure switch and to protect the instrument from damage due to cyclic pressure variations. Dampeners may be internal or external in design and shall be of adjustable type. Chemical seals shall be furnished where required.

- Process connection shall be threaded 12 mm or larger. All connection bosses shall be of substantial design to withstand repeated torque applications for leak-tight assemblies without incurring damage to the boss or injurious distortion to the instrument internal mechanism or case.
- Soft-seated over-ranged valves shall be provided to prevent excessive differential pressure from being applied in either direction of the instrument. Temperature compensation shall be provided to overcome the effect of changes of ambient temperature.

7.2.3.3 Level Switches

Level switches shall be in accordance with the following:

- Type.....External float housing type with flanged float chamber allow float removal without disconnecting piping.
- Float and Body.....Stainless steel float and steel body rated at or above the design conditions. An arrow on the float housing shall indicate the operating point of the switch. Connections 25 mm socket welded, and located on upper side and lower bottom.

All connection bosses shall be of substantial design to withstand repeated torque applications for leak tight assemblies without incurring damage to the boss or injurious distortion to the instrument internal mechanism or case.

7.2.3.4 Temperature Switches

Temperature switches shall be furnished with reference scale for set-point adjustment and shall be as follows:

Case:

- Type.....Water tight with a degree of protection of IPW 65, back flanged, front glass.

- Material and Finish .Steel, synthetic lightweight material, or equal approved by the Owner. Finish shall be dull black or gray.
- ConnectionsBottom male pipe thread and back conduit hub, both suitable sized for the service.

Measuring Element:

- Type.....Bi-metal or filled system with Bourdon tube to suit the service.
- Material.....per service temperature requirement.
- Range.....Provide nearest standard range, which is required to provide switch actuation at the maximum operating temperature for the service.

Accessories

Well shall have a 50 mm lag extension.

7.2.4 Pressure Measuring Elements

1. Bourdon tubes shall be as follow:
 - Alloy steel, preferably bored, for service of 70 bar and above
 - Steel or alloy steel, for service of 28 -70 bar
 - Bronze may be used for service of 28 bar and below

Pipe connection for directly connected instrument having:

- For dial graduation of 70 bar or greater shall be 25 mm male pipe thread.
- For dial graduation of less than 70 bar and for pneumatic receiver tube instruments, 15 mm or 8 mm male pipe thread may be supplied.

Pressure gauge for steam service shall be vacuum protected, if necessary.

2. Bellows shall be substantial leak proof design and shall be constructed of suitable corrosion resistance material.
3. Diaphragms shall be of such material that they will not harden, crack or otherwise deteriorate.

7.2.5 Liquid Level Elements

Liquid level shall be detected by floats and displacers in a chamber external to the vessel whose level is being measured.

Spherical floats shall be used for single detection e.g. for level switches. Tubular displacers shall be used to measure level over the range. Float and displacer chambers shall be designed for the pressure and temperature ratings of the vessel to which they are connected. Connection shall be flanged, screwed, or socket welded as applicable.

Provision shall be made to isolate the chambers from the vessel for maintenance purposes. Chambers shall be sealed off from the instrument mechanism housings and movement of the float or displacer shall be transmitted to the mechanism by torque tube or magnetic couplings.

7.3 Electronic Transducers and Transmitters

7.3.1 General

Transducers and transmitter shall be smart type and microprocessor based and communicating with the transmitter shall be via a hand held communicator and shall not interrupt the output of power plant. Only transducers of the low deflection type will be acceptable except that level transmitters for vacuum service shall be of the displacement type.

Transmitters shall provide a linear 4 ~ 20 mA DC output signal proportional to the process variable being measured. Transducer failure or failure of any associated component such as a pressure compensator, temperature compensator, or square root extractor shall not result in an output signal to the computer system, recorders, or indicators of greater magnitude than that representative of zero percent of the output signal range. Preference will be given to two (2) wire transmitters.

The test jack terminals shall be provided for measuring electric transducer outputs at the transducer without the necessity of disconnecting signal leads or interrupting transducer operation. For current loops, a precision resistor shall be provided in an attached enclosure with test jack or terminals. The test jacks or terminals shall be protected from environmental conditions, and be clearly and permanently tagged that they are for test purposes and indicating the test

point signal range.

If required, transducers shall contain a flexible tubing connection (metal tubing only) from the external case connection to sensing element wetted parts to minimize distortion of the transducer signal due to forces at the external piping connection.

Transducer wetted surfaces shall be suitable for the service conditions for which the transducer is furnished.

Instrument sensing piping connection may be threaded. However, they shall be 8 mm or larger. All bosses shall be of substantial design to withstand repeated torque applications for leak tight connections without incurring damage to the boss or injurious distortion to transducer internal mechanism or case. Displacement type level transducers shall have socket-welding connections.

Where welding connections are provided, the point of field connection shall terminate with a pipe extension of sufficient length to permit field welding without danger of overheating any portion of transmitter.

Shut-off and by-pass valves shall be furnished by the contractor. Transducer connections shall be spaced for standard size block valves.

Electronic dampener shall be supplied to limit amplitude of pressure variations to suit the service conditions of each transducer and to protect the transducer from damage due to cyclic pressure variations. Dampeners shall be integral in design and shall be adjustable.

In addition to normal manufacturing calibration, all transducers signal conditioners and alarm switches shall be factory calibrated for actual process operating ranges and shall be tagged with stamped metal tags stating these ranges. Material and method of attachment of the tags shall be subject to review by the Owner.

All transducers, regardless of applied service, shall be capable of withstanding their body rating conditions without damage or loss of calibration.

Differential pressure transducers, regardless of applied service, shall be capable of withstanding a differential pressure equal to full process pressure on either side of the measurement element without damage or loss of calibration.

Transducers shall be provided complete with mounting brackets. Brackets shall be of the universal mounting type.

7.3.2 Accuracy Requirements

Pressure, flow, differential pressure, and miscellaneous transducers or signal converters shall be within the standard ± 0.1 % ANSI accuracy classification and shall have repeat ability of ± 0.1 % of span or better. Zero suppression and span adjustment shall be provided to achieve maximum absolute accuracy. Transmitters shall be provided with adequate zero suppression adjustment to permit field suppression of pressure caused by the static head of the sensing line fluid. The Contractor at the time of the Bid shall furnish data specifying zero span adjustment limitations.

All instrumentation and measurement signals shall remain within ± 0.25 % of span when instrument power is supplied from its associated instrument bus. The Contractor shall provide voltage and/or frequency regulating devices, if required.

Transducers shall maintain the specified accuracies over a period of six (6) months of operation without recalibration.

Errors caused by changes in ambient temperature shall not exceed 0.01 % of span per °C change. Contractor shall explicitly state any deviation from this requirement together with data showing actual error caused by ambient temperature changes.

Transducers shall be located as near the points of measurement as possible, consistent with accessibility and avoidance of excessive temperatures, temperature variations, humidity, vibration and dust. These do exist in certain locations in the plant and their equipment shall be designed for these conditions.

Transducer cases shall be manufacturer's standard product of weather proof and dust-tight construction. Enclosure shall be of approved design with adequate accessibility to permit field-testing and calibration of transducer. Watertight enclosures with a degree of protection of IPW 66 according to IEC 60529 shall be furnished for all transducers furnished for outdoor installations and shall be completely piped and wired with transducer mounted within enclosure.

7.3.3 Electronics Transducer Requirements

Transducer outputs (except for RTDs and T/Cs) shall be DC current with AC ripple and noise reduced to 0.25 % of the maximum signal or less. Output circuits shall be completely isolated from ground (except for grounded T/Cs) and from the AC power circuit. The signal and power wires of four (4) wire transducers shall be fully segregated. Two (2) wire transducers shall be input isolated. Whenever necessary, filters shall be provided to reduce ripple and/or noise to the specific figure. Suitable shielding of the electrical circuitry shall be provided to reduce pick-up of electrical noise, which is normally high in a power generating station. Unless specified otherwise, transducer output shall be 4~20 mA DC and linearly proportional to the measured variable over its entire range.

The transmitted signal energy shall be as great as possible and circuit design shall be arranged to provide a high signal-to-noise ratio. The standard millivolt output of T/Cs is satisfactory and additional amplifiers are not required.

Transducers shall be provided with a terminal strip with screw terminals for ring lug connections to which all outgoing and test connections are made. The terminal strip shall be capable of accepting ring lug connectors using 1.5 mm² through 2.5 mm² wire. For those transducers which are locally mounted (not on an instrument panel), the terminal strip shall be furnished in a connection box which is firmly attached to or forms part of the transducer case and which provides complete segregation of the incoming cable connections from the transducer components, so that field connections may be made without exposing them to hazard normally present during installation. For four (4) wire transducers, separate conduit openings shall be provided for power wires and signal wires.

Transducers shall be provided with reverse polarity protection and current limiting. Transducers mounted outside in an open area may require lightning suppression protection, which shall be an integral part of the transducer or mounted in an attached junction box. Bidder shall provide recommendations and design details with his Bid Proposal.

Transducers shall meet all requirements with a loop impedance of 0~1000 ohm. Any change in load within the range shall not change the calibration outside the specified limits. Loop impedance greater than 1000 ohms shall not

damage the transducer, its power supply, or any other instrument outside the specified limits.

7.3.4 Flow Transducers

Flow transducers shall measure the differential pressure developed across a primary flow element and transmit electrical output signals for use by the centralized control, display and plant computer system or the local control and indication systems as required.

For the centralized system the output signals shall be proportional to the square of the flow. The centralized system will extract the square root and correct the signal for variations in temperature and pressure of the process.

For the local control and indication systems, the output signals may be proportional to the square root of the flow or be linearized within the transducer as required.

The transducers shall have capability to provide any necessary corrections for ambient temperature variations.

Electro-magnetic flow meter shall be provided with pulsed DC excitation. The flow meter will be preferably SMART type and shall be provided with a hand held calibrator for ease of instrument calibration. Accuracy (including combined effects of linearity, hysteresis and repeatability) shall be $\pm 0.5\%$ of calibrated span or better.

Turbine flow meters (if required) shall be provided with integral signal converters. Outputs will be 4 – 20mA DC isolated. Accuracy (including combined effects of linearity, hysteresis and repeatability) shall be $\pm 0.5\%$ of calibrated span or better.

Positive displacement meter will preferably be oval gear type with accuracy (including combined effects of linearity, hysteresis and repeatability) of $\pm 0.5\%$ of calibrated span or better and a local flow integrator shall be provided. If the instrument is provided with transmitter output, the same shall be 4 – 20mA DC isolated two-wire type.

Integrator shall be mounted in the transmitter, cases when specified and shall be of the odometer type, visible without opening the door of the case.

7.3.5 Pressure and Differential Pressure Transducers

Unless specified otherwise pressure and differential pressure transmitters shall have a minimum zero suppression range of 25% of span.

Transducers for measuring absolute pressure shall utilize an evacuated chamber whereby the measurement is not subject to change in barometric pressure. Transducers in this category must withstand the maximum pressure as stated in the Instrument Data Sheet or up to four (4) times full- scale pressure (whichever is larger) without permanent damage or loss of calibration. The accuracy shall not less than 0.5%.

7.3.6 Level Transducers

Output signals from level transducers shall be directly proportional to level measurements unless specified otherwise.

Reservoirs (constant head chambers) shall be furnished for all level transducers of the differential pressure type used on pressurized vessels. Inlet piping connection shall be 20 mm minimum, socket-welded, and outlet piping connections shall 15 mm minimum, socket welded, both suitable for the pressure and the temperature listed on the Instrument Data Sheet. Reservoir material shall be suitable for the process conditions.

Differential-pressure level transducers used in conjunction with reference chambers or legs shall be capable of zero suppression approximately equal to their differential pressure ranges and shall be capable of withstanding full vessel pressure on either measuring element (with the other element at atmospheric pressure) without damage or loss of calibration.

Level transducers for vacuum service shall be of the displacement type and shall be provided with 50 mm socket-welded connections. The accuracy shall not less than 0.5%.

7.4 Pneumatic Transmitters

Transmitters of the pneumatic type shall operate with an air loading pressure of range 3-15 psi.

Transmitters shall be equipped with a “Supply Air” and a “Transmitted Air” gauge.

Transmitter output shall be linear with operating variable being measured.
Transmitters shall include an air set with filter and regulator.

7.5 Local Indicators

7.5.1 Pressure Indicators

The pressure indicators furnished shall be in accordance with the following unless specified otherwise:

Type Bourdon tube, diaphragm, or bellows type based on service for which the pressure indicator are used.

C a s e :

Design..... Weather proof

Type..... Glass front with gasket

Material..... Steel, synthetic lightweight material or equal approved Owner

Connection..... Threaded and appropriate size for pressure range required per service, bottom connected

D i a l

Size 150 mm

Color White dial with black letters, numerals, graduations, and Pointers

Pointers Equipped with a mechanism for minor adjustment

Stop pins Clearly inscribed on the dial face for the service for which the instrument is intended, as indicated on the Instrument Data Sheets. Tape, metal tags, or plate are nor acceptable

Accuracy Within ± 1 % of span

Range Provide nearest standard range above one required for maximum operating condition

Accessories:

All pressure indicators on steam service, shall be complete with siphons. Other

accessories such as vibration dampers and chemical seals required to suit the service shall also be furnished.

7.5.2 **Thermometers and Thermo-wells**

The thermo meters furnished shall be in accordance with the following unless specified otherwise:

Type Bi-metal, dial type with adjustable face with every-angle construction. Neck to be capable of 360°C rotation and stem capable of being turned 180°C without damaging the instrument.

C a s e

Design Weather proof

Type Glass front with gasket

Material Steel synthetic lightweight material or equal approved by Owner

Connection Manufacturer's standard threaded size

D i a l

Size 150 mm

Color White dial with black letters, numerals, graduations, and pointers

Pointers Equipped with a mechanism for minor adjustment

Stop pins Clearly inscribed on the dial face for the service for which the instrument is intended, as indicated on the Instrument Data Sheets. Tape, metal tags, or plate are not acceptable

Accuracy Within ± 1 % of span

Range Provide nearest standard range above one required for maximum operating condition

Stem Stainless steel with welded joints. Length shall be such as to match the associated well

Accessories:

Wells shall have a 50 mm lag extension.

7.5.3 Level Gauges

The Contractor shall furnish and install gauge glass on each vessel or tank where specified.

Gauge glasses shall be in accordance with the requirements set forth herein and in the specification for vessel or tank to which they are connected. In case of conflict, the specification for the tank or vessel shall override this specification.

Each gauge glass shall be of such total length as to effectively cover the water level range of the vessel or tank to which it is connected. Where the level range is too great for a single gauge glass, multiple gauge glasses in close proximity to each other shall be provided. They shall be located in such a manner that the water level is always visible.

Gauge glasses shall be of transparent glass safety type with mica shield on the inside of the glass as approved by the Owner. They shall be suitable for the vessel or tank shell design pressure at saturated temperature as follow:

- | | |
|-------------------|------------------|
| - 50 bar ~ 70 bar | -Series19-TM-40 |
| - 25 bar ~ 50 bar | -Series19-TM-30 |
| - below 25 bar | -Series 19-TM-30 |

or equal approved by the Owner.

Spring washers shall be used between the cover and nuts on Series 19-TM- 40 gauge glasses.

Each gauge glass shall be inspected in the field and the gauge cover bolts tightened before being placed in service. Each gauge shall be provided with a metal tag securely attached to the gauge at the factory indicating clearly the correct torque that shall be applied to the nuts when tightening in the field to ensure proper bolt tension.

Each gauge glass assembly shall be provided with 12 mm socket-welding connections. Pipe, fittings, and shut-off valves between the tank, heater shell or hot well and the gauge glass assemblies shall be furnished and installed by the Contractor.

7.6 Panel Mounted Instruments

7.6.1 Indicators

7.6.1.1 Cases

- a. Cases shall be flush mounted, back connected and of synthetic lightweight material with dull black finish. Bezels shall be removable screwed or hinged type with a gasket for tight closure. Dials shall be accessible from the front of the indicator. Cover glass shall be manufacturer's standard clear type unless other wise specified.
- b. Circular indicator cases shall be of 110 mm nominal diameter.
- c. Vertical scale indicator cases shall be nominally 180 mm high and width to unit the number of indicator units.
- d. Square indicator cases shall be nominally 110 mm by 110 mm.

7.6.1.2 Dials and Scales

- a. Dial and scales shall be white back-ground with black letters, numerals, and graduations, and shall contain service identification.
- b. Letters for service identification shall be black for dials and single indicating scales and the left hand identification of double indication scales. Letters for the right hand identification of double indication scales shall be red.

7.6.1.3 Pointers

- a. Pointers shall be black for dials and for single indicating scales and for the left hand indication of double indicating scales. Pointers shall be red for right hand indication of double indicating scales.
- b. Dial pointer hubs shall be of steel or other hardened metal as approved.
- c. Pointers shall have micrometer adjustment.
- d. Maximum deflection of Ammeters shall be at least 6 times nominal current.

7.6.1.4 Movements

Movement shall be stainless steel with nylon pinion, or equal approved by the Owner.

7.6.1.5 Temperature Indicators

Temperature indicators, which receive signals directly from T/Cs, shall be equipped with automatic cold junction temperature compensation.

7.6.2 Recorders

7.6.2.1 Cases

- a. Cases shall be flush mounted, dull black finish, back connected with hinged door for access to chart and mechanism from the front of instrument. The door shall be gasketed for tight closure and door handles shall include a built in lock. Cover glass shall be manufacturer's standard clear type.
- b. Each case shall have a legend plate fastened to the inside of the door to carry the color designation for multiple point or pen recorders, the recorder's instrument number and the service description.

7.6.2.2 Chart Drives

- a. Chart drives shall be arranged to operate satisfactorily on a 220 V, 50 Hz, single phase, AC circuit.
- b. Recorders shall be servo-potentiometer type, internally illuminated and shall be complete with lamp and internal wiring for lamp and motor. They shall have 600 mm extension cords and polarized twist lock type rubber plugs for connection to the electrical supply.
- c. Alarm contacts shall be provided, if required. Contacts shall be rated for 0.5 Amps 250 V DC inductive and shall close for alarm condition. Alarm contacts on multipoint recorders shall remain latched while the recorder is scanning other points. Relays shall be furnished if necessary and shall operate on the same source as the chart drive.

7.6.2.3 Strip Charts

- a. Charts shall have 12 midnight sur-stamping repeated leaving blank space for tear-off.
- b. Strip chart instruments shall be equipped with tear-off device and setting indices to re-time the chart at daily tear-off.
- c. Strip chart instruments shall operate at a speed of 25 mm per hour, unless specified otherwise.

- d. Strip charts shall be the folded type.

7.6.2.4 Pens

Pen arms shall operate on pivoted centers and inking systems shall be designed to provide legible record without excess inking.

7.6.2.5 Multipoint Records

Multi point instruments shall have records consisting of multi colored printing only, no numbering of point being included. Adjacent point on charts shall have the greatest color contrast.

7.6.2.6 Scanning Records

Scanning type instrument shall print mono-colored record of any and all points.

7.6.2.7 Chart Supply

400 circular charts or one year's supply of strip chart and ink, or inked pad shall be furnished with each instrument. Stamps and stamp pads shall be included, giving legend plate information, unit number and data line. Supplies shall suitably boxed and identified.

7.6.2.8 Temperature Recorders

Temperature recorders, which receive signals directly from T/Cs, shall be equipped with automatic cold junction compensation.

7.6.3 Alarm Annunciators

For alarm annunciator, refer to Part 4, Technical Specification I-002 "Local Control Panels".

7.6.4 Push Buttons, Control Switches, and Indicating Lights

For push buttons, control switches and indicating lights, refer to Part 4, Technical Specification E-001 "General Electrical Specification".

7.6.5 Totalizers

Totalizers shall be suitable for flush mounting. Bezels shall be finished matt black.

Totalizers shall be miniature, solid-state type with a minimum of seven figures of brilliant LED display and a front of instrument-reset push button.

Totalizers shall accept count inputs from standard source including contact closure at a minimum rate of 100 closures per second.

7.7 Special Measurement and Equipment

7.7.1 Vibration Measurement and Analysis

7.7.1.1 General

Measurement and monitoring of vibration shall be made on all large rotating or reciprocating machinery including gas turbine/generator, steam turbine/generator and large motors for protection and predictive maintenance.

Suitable indicators with brightly colored fluorescent barograph display shall be supplied in the relevant Control Panels for each measurement points and the measurements shall be suitably alarmed where high vibration level may cause possible damage or affect the safety of the plant. An on-line Data Acquisition System and software on a Personal Computer shall be provided for vibration data trending, FFT analysis, etc. The system shall be able to accept other analog inputs e.g. temperature, pressure, etc. The basic vibration data shall also be transferred to the DCS for display to the operator.

Routine report and hardcopies required are Trend, Alarm Polar Plot, FFT Spectrum, Time Domain Wave form, Run-up Run-down Bode plot, etc. The speed of FFT shall be typically 0.25 second for 16 simultaneous channels. Training on the complete system shall be provided.

For shafts on journal bearings, two (2) proximity probes shall be furnished for shaft radial vibration measurements located at 90 degree apart radially. Where it is not possible to install proximity probes, two (2) accelerometers on the casing in two (2) directions shall be provided at each bearing. A key phasor is also to be provided. For shaft axial position for the steam turbine, two (2) proximity probes shall be mounted parallel to the axis of the shaft.

For shaft on anti-friction bearings and gearboxes, two (2) accelerometers on the casing in two (2) direction shall be provided on each bearing. The frequency response on the accelerometer shall be able to cover the bearing and gear mesh frequencies.

The output of each sensor shall be based on current modulation technique and be made available to the Plant Management Station in the Central Control Room.

All enclosures in open areas shall be watertight with degree of protection of IP 66 according to IEC 60529.

7.7.1.2 Non Contacting Type

Non-Contacting type vibration measuring devices shall include proximity probes, transmitter and interconnecting cables. The proximity probe's sensitivity shall be temperature compensated to $\pm 5\%$ throughout its temperature range. Front-end transducer cables shall be protected with flexible stainless steel tubing.

To ensure that the output of the probes is truly representative of shaft vibration, the shaft shall be free from surface irregularities and have a finish which is concentric, homogeneous, demagnetized, and have a run-out of less than $2\text{ }\mu\text{m}$ peak-to-peak.

This shaft condition shall extend over a surface area that is equivalent to the full float of the machine and to a depth greater than the magnetic field penetration (approx. 0.38 mm). The shaft shall not be chrome plated. The shaft vibration measurement shall be according to ISO 7919.

7.7.1.3 Contacting Type

Contacting Type vibration measuring devices shall include acceleration sensors as required, amplifiers, switchable integrators, rectification devices, etc. and interconnecting cables. All transducers shall meet the temperature and frequency response requirements. There shall be galvanic separation of to 4 kV between transducer and monitoring electronic to eliminate earth loop problems.

7.7.1.4 Output and Power Supplies

Each sensor, cable and transmitter combination shall form a matched compatible unit to ensure that the transmitter output is representative of the vibration measurement.

Each transmitter output shall be of varying 4~20 mA DC signal containing amplitude and frequency information about the vibration.

Galvanic isolation shall be provided between vibration sensor and monitoring electronics to eliminate potential earth loop problem due to different potential between control room and machine.

7.7.1.5 Installation

Proximity probes shall be rigidly mounted to the reference structure, (bearing housing, or machine case) and shall have a clear field of at least 3 x tip diameter. The probe position shall be as close to the bearing housing as possible within the limits of 3 x tip diameter if it is not possible to mount it at the center of housing. The probe shall be withdrawable while the machine is running.

Contacting transducers shall be positioned horizontally on a machined pad on the bearing housing wherever possible and shall be securely mounted.

7.7.1.6 Prefabricated Plug-In Interconnecting Cables

When applicable prefabricated plug-in interconnecting cables shall meet the following requirements:

- a. The interconnecting plug-connected cables shall be furnished complete with plugs and receptacles furnished and wired at both ends.
- b. Insulation shall be high quality XLPE (Cross linked Poly Ethylene) of the type that does not support combustion. The conductors shall be grouped with a 1.5 mm², stranded, tinned coated, copper drain wire and shielded with aluminum and mylar tape with aluminum in continuous contact with the copper drain wire and shall have an overall jacket of flame-retardant material
- c. Plug shall be military type.
- d. Each cable shall be tagged at both ends with a metallic or non flammable cable tag with identifying number.

7.7.2 Chemical Analyzers

The Bidder shall provide technical data relating to each type of analyzer in his offer, including but not limited to:

- (i) Specification of physical and chemical conditioning of sample gas stream and water.
- (ii) System requirements of accuracy and chemical conditioning of sample gas stream and water.
- (iii) Details of analysis instrument.
- (iv) Display and output signal level information

Single sample analyzers shall be used, except in cases where the Owner has specified or has given written agreement to multi sample types.

Wherever possible analyzer shall utilize an analyzer probe.

The use of individual or multiple prefabricated analyzer installation shall be used to reduce Site installation work. This prefabrication shall include sample conditioners, analyzers, air and electrical distribution cooling water distribution or coolant circulating system all piped and wired on a common frame. The arrangement shall permit testing of the entire assembly before dispatch to site and shall be arranged for convenient removal from on-line operation to facilitate routine maintenance and calibration.

Generally analyzers shall be suitable for mounting in ambient plant conditions, however when this is not feasible e.g. spectrographic or chromatographic analyzers, they shall be located in an air conditioned room.

A schedule of recurring maintenance and adjustment operations shall be prepared and the necessary environment for these operations specified.

Flammable vapour effluents, including sample discharge lines shall be disposed to an atmospheric vent which is independent of any process vent. The vent shall be taken to a safe height and its location agreed with the Owner.

The sampling system shall include but not be limited to all probes, valves, filters, coolers, pressure regulations, coalescers, flow meters, piping and pumps as necessary, to give the analyzer a representative and suitable conditioned sample.

The delay time between a sample being withdrawn from the process and being available to the analyzer shall be consistent with the measurement and control requirements of the particular process. In all control applications (including

alarm and protection) the permissible measurements lag shall be agreed with the Owner. The preferred method of keeping sampling lags to acceptable limits shall be by locating the analyzer close to its sample take-off point.

When this is not practicable, the distance-velocity lag of the sampling system shall be reduced by increasing the velocity of the sample, e.g. by fast loop. In such cases the design calculations shall be submitted to the Owner for approval. The sample shall be returned to the process or disposed of in an economic and safe manner.

Data from the analyzer shall also be available to the Plant Management Station for monitoring and data archiving.

7.7.2.1 Fuel Gas Chromatograph (not applicable)

On line fuel gas chromatograph shall be supplied for fuel gas analysis specified in M 414 “ Fuel Gas System “.

- Measuring item shall be CO₂, N₂, CH₄, N₂S specific gravity, gross heating value, etc.
- Analyzer shall be explosion proof type
- Analyzer signals shall be transmitted and monitored to Plant Management Station.
- Electrical power supply for analyzer shall be 220 VDC or 220 VAC.

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7.7.2.2 Steam and Water Analyzer

All measuring signals for steam and water analyzer specified in M-400, M-401, M-404 and M-406, shall be transmitted and monitored to Plant Management Station.

Electrical power supply for analyzer shall be 220 VDC or 220 VAC

Measuring signals shall be 4~20 mA DC

7.7.3 Continuous Emission Monitoring System (CEMS)

Measuring signals for flue gas monitor specified in M-301 and M-305 shall be transmitted and monitored to the Plant Management System.

Electrical power supply for analyzer shall be 220 VDC or 220 VAC.

Measuring signals shall be 4~20 mA DC

The measuring range for these instruments shall be followings;

- CO : 0 ~ 100 ppm / 0 ~ 1000 ppm
- CO₂ : 0 ~ 5 Vol % / 0 ~ 20 Vol %
- SO₂ : 0 ~ 100 ppm / 0 ~ 1000 ppm
- NO_x : 0 ~ 100 ppm / 0 ~ 1000 ppm
- O₂ : 0 ~ 10 Vol % / 0 ~ 25 Vol %

7.8 Actuator

7.8.1 General

Unless otherwise specified actuators for modulating valves and dampers shall be pneumatic. Self contained sealed hydraulic units may be considered where high thrusts or high speeds of operation are required but each application shall be to the approval of the Owner.

Actuators for ON/OFF duty or manually positioned units shall be electrical motor driven, however, the use of solenoid types on small valves may be allowed dependent on duty.

The operation of all actuators, control valves and driven units, shall be arranged to ensure the safety of the plant under failure of control or actuating supplies. Unless specified otherwise, the failure of a control signal or actuator power supply shall either:

- cause the actuator to move to a safe position
- freeze in its last operating position

With either action the failure mode shall be suitably monitored and the plant operator informed by some form of alarm.

Where pneumatic actuators are used, an internal bias spring shall be used to obtain motive power to reach the safe position in the event of supply failure.

Where the type of actuator offered does not have the appropriate fail-safe facilities described above then the valve/damper shall have a second valve/damper in series or parallel as appropriate, designed to provide the

correct failure response.

When an actuator is locked or moves to a safe position on failure detection, it shall not be allowed to return to automatic control without resetting action by operator after restoration of the supply/control signal. Similarly remote manual control should not be allowed to return automatically if this would allow the possibility of large disturbances to be injected into an operating plant.

The failure response to all actuators in the event of the loss of the prime mover (air/oil pressure, electrical power) shall be indicated on the flow diagrams and others, valve schedules, etc.

Where direct position control is provided on a remote control panel an actuator position feedback signal and indication shall be provided which is independent of the position demand signal.

Where control requirements call for split range operation, the computing of the split range shall be carried out externally from the actuator such that standard signal inputs to the actuator are maintained.

All actuators shall include a mechanical device to show the true position of the operating system.

All actuators shall have a handwheel for direct manual operation. The diameter of the handwheel and geared effort shall be such that they are reasonably operable by one man. A lockable mechanical clutch mechanism shall be provided to inhibit power control of the actuators when the handwheel is being operated. The disengagement for hand control shall be signalled remotely to indicate that the normal operation of the actuator is inhibited.

All actuators shall be provided with a local control facility which in general will be used for test purposes only. Such control may take the following forms, as appropriate or specified.

- Control initiations (i.e. raise, lower, etc.) with lockable local/remote selection when appropriate either on the actuator, in the direct vicinity of the actuator or on its associated switchgear.
- A portable test facility for injecting the appropriate position demand signal either at the actuator or drive unit (switchgear or power amplifier).

7.8.2 Electrical Actuation for Isolating and Binary Control

Unless specified, integral or separate switch and actuator units may be offered for which the contactors may be of the mechanical or static switch type. If a static switch is offered then the switch unit and actuator shall be supplied by the same manufacturer as a complete system.

The following features shall be provided for each actuator

- DC open and close (raise and lower) command interposing relays.
- Open and close travel limit switches (each with change over contacts).
- Open and close indicating lamps.
- Instantaneous voltage phase sequence/failure monitoring relay.
- Padlockable local/remote control selector switch.
- Analogue actuators position signal shall only be supplied when specified.
- Thermal overload protection.
- Torque limit switches.

The DC interposing relays (required above) shall be energized by the I & C system power supplies and shall therefore be rated to be compliant with the standard binary signal levels (48V, 24V or other standard agreed voltage).

All switch units & actuators shall be incorporated in enclosures with a protection class of at least IP54. Better type enclosures shall be provided where specified for special applications. The main control and contactors shall be incorporated directly on the actuator mechanism. Contactors, if not integral with the actuator, shall be incorporated in the actuator boards, in individual compartments logically arranged to their function.

7.8.3 Pneumatic Actuators

Pneumatic actuators shall have the following attributes:

- They shall cause the valve or driven unit to operate over its full stroke, from a 0.2 bar to 1.0 bar gator
- Pneumatic signal to a positioner
- Electronic signal to electro/pneumatic convertor

- Electrical signal to solenoid pilot valve

Pneumatic actuator and drive unit shall automatically return to the rest position upon signal and/or air supply failure unless 'process control' or the specification requirements dictate for 'stay-put' response.

Where lock up devices are provided on those services requiring the actuator to remain in the position prevailing immediately before an operating medium failure, the following two methods of "freezing" are acceptable.

- The pneumatic system may be locked, but equipment used for this purposes shall always be installed between a positioner output and an actuator input.
- The final regulator may be locked by mechanical means upon receipt of a failing supply pressure signal.

All pneumatic control equipment, control drives and control valves, shall be capable of satisfactory operation on a main air supply pressure of approx. 7 bar g normal down to 5.5. bar g minimum.

Positioners / actuators accepting standing or pulsed position reference signals are equally acceptable.

Positioners shall be used on the following applications.

- When the valve is 100 mm or larger
- When trouble may be expected from an unbalanced valve
- When there is a large time lag in the controlled loop (temp.)

7.8.4 Electro Hydraulic Actuators

When applicable prefabricated plug-in interconnecting cables shall meet the following requirements:

- a. The interconnecting plug-connected cables shall be furnished complete with plugs and receptacles furnished and wired at both ends.
- b. Insulation shall be high quality XLPE (cross linked poly ethylene) of the type that does not support combustion. The conductors shall be grouped with a 1.5 mm², stranded, tinned coated, copper drawing wire and shielded with aluminum and mylar tape with aluminum in continuous contact with the copper drain wire and shall have an overall jacket of

flame-retardant material.

- c. Plug shall be military type.
- d. Each cable shall be tagged at both ends with a metallic or non flammable cable tag with identifying number.

7.9 Air Supply System

Where pneumatic actuators are provided two sets of air filter and pressure reducing equipment shall be provided to supply air to a header on which valves shall be fitted to enable individual instruments to be isolated. Headers/systems supplying air at reduced pressure directly to the instrument shall be provided with a safety relief valve appropriate to the set pressure. The header air supply “take- off” connections shall be from the top of the pipework. Lower elevation points shall be fitted with drain valves.

All instrument air supply piping shall be of seamless steel electro- galvanized construction as a minimum requirement, with the application of a ring main system as a preference.

Piping fittings shall also be of steel and electro-galvanized.

For reasons of mechanical strength, the air supply ring main and headers on pipe tracks shall have a minimum size of 2 inches N.B. or equivalent.

The size of the distribution pipes shall depend on the number of users along with their individual consumption and shall conform to the following as a general requirement.

<u>Number of Users</u>	<u>Pipe Sizes</u>
1 - 5	½ in.N.B
6 – 20	1 in. N.B
21 - 50	1 1/2 in. N.B
51 - 100	2 in. N.B
101 -200	3 in. N.B

Pipes sizes of metric equivalent are acceptable.

At least 15% spare branches 1/2 in. N.B or equal shall be provided, evenly distributed through the plant area.

All distribution valves and individual isolating valves shall be preferably of bronze or brass construction globe type, unless otherwise agreed by the Owner.

The use of underground air supply lines shall only be by agreement and discretion of the Owner.

The route(s) selected for pneumatic signal lines shall be as straight as possible.

Pneumatic signal tubing shall be of 6.0 mm (metric) or 0.25 ins (Imperial) minimum O.D. with wall thickness of 20 SWC or equivalent.

Tubing to individual instruments; in the field, shall be in copper and sheathed in PVC, unless otherwise agreed.

All compression fittings shall be of an approved manufacture.

All tubing shall be neatly run and adequately supported on brackets and tray runs. Ends of tubes shall be identified by permanent ferruling.

7.10 Installation

All instrument installations shall be carried out in accordance with recognized and approved standards and shall comply with the manufacturer's instructions.

All instrumentation shall have isolation and where appropriate, drains and venting facilities to permit removal and re-installation of all instruments whilst the plant is in operation.

All necessary accessories such as root valves condensing chambers dirt traps chemical seals etc for instrumentation shall be provided.

All instruments shall have double isolation for high pressure application and, where appropriate, venting facilities to permit removal of all instruments whilst the plant is in operation.

All pressure instruments shall be installed vertically and means shall be provided for venting pressure from the line and instrument so that it may be removed safely.

Adequate test points for portable instruments shall be provided for the testing and maintenance of all remote and local measurement loops.

Gas or liquid lines shall not be routed into instrument panels, such measurements shall be provided by locally mounted indicating transmitters.

Instruments mounted in the field shall not be supported from pipework, stairways or railings.

Recorders, transmitters, controllers, pressure switches and like shall be supported independently of the pressure connection. The type of support shall depend upon the make of instrument; care must be taken to avoid the possibility of imposing stresses from the pressure piping, conduit etc. which may cause malfunction.

Drip trays or effective barriers shall be provided to prevent damage to electrical equipment due to fluid leakage.

Instruments shall be adequately protected to ensure long term service in the operating environment and shall be mounted in such a manner as to avoid the effects of vibration, extremes of temperature, dust, humidity and salt spray etc. Instruments shall not be located beneath tanks and vessels and other areas where spillage or overflow will cause the fluid to come into contact with the instrument.

The instrument installation shall make due allowance for thermal cladding. The length of temperature sheaths, impulse piping etc. shall be selected to prevent disturbance of cladding during instrument removal.

The Bidder shall include with his offer typical hook up drawings for each type of measurement he proposes to use on this project.

8. TESTING

8.1 Test Facilities

Equipment furnished under this Technical Specification shall be designed and arranged to facilitate periodic testing, calibration and maintenance.

8.2 Test Equipment

Special test equipment and standard test equipment required to adjust, test or maintain any equipment furnished by the manufacturer contractor shall be provided by the Contractor.

8.3 Laboratory and Factory Testing

Laboratory and Factory Testing shall be performed as **IEC or ANSI standard.**

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8.4 Performance Test

After equipment is put into service, the Contractor shall run a performance test to ascertain that the equipment is performing properly in every respect. This shall be in addition to tests performed at laboratories and Contractor's factory prior to shipment. The Contractor shall remedy any deficiencies identified at no extra cost to the Owner.

9. NAMEPLATES

Nameplates shall be provided as described in Tech. Specification I-002 "Local Control Panels".

10. PACKAGING INSTRUCTIONS AND STORAGE

The Contractor shall fully comply with all requirements of Part 4, Sub-section 2.12 and in addition shall shop finish paint all instrument black or dark grey.

11. QUALITY ASSURANCE / QUALITY CONTROL (QA/QC)

The Contractor shall conform to the quality assurance / quality control requirements of Part 4, Sub-section 2.7, clause 4 unless the mechanical equipment with which the Instrumentation and Controls are associated is Level 1 in which case Clause 3 shall be applicable.

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